# The Effective-Orbit One-Bit Flip Chain of a Frozen Hénon Support Core

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**Abstract**

We quotient the sixteen-dimensional support cube by the faithful finite label action of order 1920. The resulting one-bit flip chain has 3024 states, constant row sum 16, and 30240 nonzero directed entries. We certify strong lumpability at every support, orbit-size detailed balance, the exact flow between four repair-distance levels, and the complete spectrum on orbit-constant functions. Its eigenvalues are $16 - 2k$, with multiplicity equal to the number of label-group orbits on $k$-subsets. The repair-zero diagonal flow is 445696, independently reproducing the distance-one correlation from C82. The result is finite and combinatorial under the firewall `NO_BAD_EULER_OR_ROOT_NUMBER`.

## 1 Finite action and quotient

Let $L$ be the frozen set of sixteen named labels and identify a retained support $A \subseteq L$ with a vertex of $X = \{0,1\}^{16}$. C75 supplies an ambient lifted action of order 11520, whose kernel on labels has order 6. Accordingly, every support calculation here uses the faithful image $E \leq \mathfrak{S}_L$ of order 1920, never the ambient order as though it were faithful. The action has 3024 orbits on $X$.

Write $\mathcal{O}_i$ for these support orbits. For $A \in \mathcal{O}_i$ define

$$q_{ij} = \#\{\ell \in L : A \mathbin{\triangle} \{\ell\} \in \mathcal{O}_j\}.$$

Thus $Q = (q_{ij})$ records how the ordinary cube adjacency operator acts on functions that are constant on $E$-orbits.

**Theorem 1** (Exact strong quotient)**.** *The number $q_{ij}$ is independent of the chosen $A \in \mathcal{O}_i$. Every row of $Q$ sums to* 16*, and*

$$|\mathcal{O}_i| q_{ij} = |\mathcal{O}_j| q_{ji}.$$

*Consequently the orbit-size measure is reversible. The quotient has* 30240 *nonzero directed entries and* 15120 *unoriented orbit pairs.*

*Proof.* For $g \in E$, toggling is equivariant: $g(A \mathbin{\triangle} \{\ell\}) = gA \mathbin{\triangle} \{g\ell\}$. Since $g$ permutes the sixteen labels, it bijects the toggles from any two representatives of $\mathcal{O}_i$ and preserves target orbits. This proves strong lumpability and the row sum. Counting actual cube edges from $\mathcal{O}_i$ to $\mathcal{O}_j$ in the two orientations gives the detailed-balance identity. Support-size parity changes under every toggle, so there are no self-loops; hence directed entries occur in opposite pairs. □

The number of distinct target orbits in a row has distribution

| targets | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
|---|---|---|---|---|---|---|---|
| rows | 128 | 384 | 480 | 800 | 864 | 336 | 32 |

and positive entries $1, 2, 3, 4, 5$ occur respectively $19296, 7344, 1008, 1584, 1008$ times.

## 2 Complete invariant spectrum

For $B \subseteq L$, let $\chi_B(A) = (-1)^{|A \cap B|}$ be the Walsh character. A direct cube calculation gives adjacency eigenvalue $16 - 2|B|$.

**Theorem 2** (Invariant Walsh spectrum)**.** *The complete spectrum of $Q$ is $16-2k$, $0 \le k \le 16$. Its multiplicity is the number of $E$-orbits on $k$-subsets, namely*

| $k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $m_k$ | 1 | 7 | 27 | 73 | 151 | 252 | 352 | 424 | 450 |
| $k$ | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | |
| $m_k$ | 424 | 352 | 252 | 151 | 73 | 27 | 7 | 1 | |

*The multiplicities sum to* 3024*; the first and second spectral moments are* 0 *and* 77760*.*

*Proof.* The Walsh characters form an orthogonal eigenbasis of functions on $X$. The action of $E$ permutes them by $g\chi_B = \chi_{gB}$. Summing characters over each orbit of subsets produces a nonzero invariant eigenfunction. Distinct orbit sums have disjoint Walsh support and are independent; every invariant function has coefficients constant on those orbits. They therefore form a basis of the invariant subspace, which is identified with functions on the 3024 support orbits. The stated multiplicities follow. □

# 3 Repair flow and predecessor identity

Let $\rho(A) \in \{0, 1, 2, 3\}$ be the frozen repair distance from C78. The numbers of support orbits at these four levels are $1332, 1500, 180, 12$. Counting actual directed cube edges by the pair of repair levels gives

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 445696 | 40704 | 0 | 0 |
| 1 | 40704 | 469376 | 13184 | 0 |
| 2 | 0 | 13184 | 24064 | 640 |
| 3 | 0 | 0 | 640 | 384 |

whose entries sum to $16 \cdot 2^{16}$. In particular, the $(0, 0)$ entry is the number of ordered Hamming-distance-one pairs for which both endpoints have the full core. Its value 445696 exactly equals the C82 distance-one autocorrelation, providing a cross-paper identity obtained by a different aggregation route.

# 4 Certificate and scope

The producer explicitly generates all 1920 label permutations, whereas the independent checker constructs support-orbit components using only five named generators. Both verify every one of the 65536 supports and every quotient row. A symbolic radial-cube check, clean replay, and twenty hostile mutations complete the audit. The canonical evidence SHA-256 is

`7b3e2179590c3dc8662a59f1d79ffbb1`
`2f2a4a787438a6902d6c28b2842e70b8`.

No arithmetic/local data, Euler factors, root numbers, automorphy, full Burnside ring or table of marks, or Hilbert–Polya operator is claimed.